



TANG HUI YI

Data Analyst

+601123191627 | hytang0806@gmail.com | Pulau Pinang, Malaysia

PROFESSIONAL SUMMARY

Final-year **Decision Mathematics & Data Analytics** student (CGPA 3.69/4.00) with hands-on experience in **Python, SQL, Tableau, Machine Learning, and Data Visualization**. Built end-to-end analytics projects involving customer churn prediction, car price forecasting, and operational simulations using real-world datasets. Passionate about transforming data into actionable business insights and seeking a Data Analyst Internship opportunity. Available for internship from **August 2026 to November 2026**, with flexibility for extension based on organizational requirements.

TECHNICAL SKILLS

Programming	Python, SQL, Java
Data Analytics	Tableau, Excel, Streamlit, Orange Data Mining, Power BI
Machine Learning	Logistic Regression, Random Forest, Decision Tree, Classification Modeling
Data Skills	Data Cleaning, EDA, Data Visualization, Feature Engineering, Model Comparison
Operations Research	Simulation Modeling, Simulated Annealing, Optimization

EDUCATION

Universiti Utara Malaysia (UUM)	Sep 2023 – Nov 2027
<i>Bachelor of Decision Mathematics & Data Analytics CGPA: 3.69/4.00 Minor: Information Technology</i>	
SMJK Jit Sin	Aug 2021 – Mac 2023
<i>Sijil Tinggi Persekolahan Malaysia (STPM) Science Stream - PA, MathT, ICT, Physics</i>	
SMJK Jit Sin	Jan 2016 – Feb 2020
<i>Sijil Pelajaran Malaysia (SPM) Science Stream</i>	

PROJECT EXPERIENCE

Customer Churn Prediction (Telecommunications)

Python | Machine Learning | Streamlit

- Analyzed 7,043 customer records with 21 variables from the IBM Telco Customer Churn dataset.
- Performed data cleaning, feature engineering, encoding, and class imbalance treatment.
- Evaluated six classification models including Decision Tree, Random Forest, and Logistic Regression.
- Achieved over **80% prediction accuracy** using Logistic Regression as the final production model.
- Developed and deployed an interactive Streamlit dashboard enabling real-time and batch churn risk prediction.

Car Price Prediction System

Python | Selenium | Random Forest | Streamlit

- Scraped **3,480+ used vehicle listings** from Mudah.my across 88 pages using Selenium WebDriver.
- Cleaned and transformed raw pricing and mileage data for model training.
- Developed a Random Forest model incorporating vehicle location, age, and mileage as key variables.
- Built a Streamlit application allowing users to estimate vehicle market prices instantly.

Simulated Annealing Optimization

Java | Optimization

- Developed a Simulated Annealing algorithm to solve a 100-agent Linear Assignment optimization problem.
- Reduced total assignment cost from **4,828 to 360**.
- Achieved 92.5% optimization improvement within 14 seconds.

EMPLOYMENT EXPERIENCE

Event Operations & Support Crew | MIMAC & Petrina Cup

Sep 2023 - Nov 2025

- Coordinated event logistics and crowd management for large-scale university events.
- Resolved operational issues on the spot under tight timelines.

After-School Tutor | Pusat Tuisyen Cemerlang Impian

Aug 2025 - Oct 2025

- Taught students across multiple subjects and adjusted teaching methods based on each student's learning pace and needs.
- Managed time across concurrent classes and tracked individual student progress to ensure consistent improvement.

Inventory Assistant | Textran Industries (Penang) Sdn. Bhd.

Jun 2021 - Sep 2021

- Processed customer orders, maintained real-time stock records, and updated inventory data to ensure accurate tracking of outgoing products.

LEADERSHIP & ACTIVITIES

Secretary | 7dian9 Song Composing Concert 2025, UUM

Oct 2024 – April 2025

- Led cross-department coordination across multiple teams at the same time, managing communication, meeting records, and university approvals for a large-scale student concert.
- Handled field responsibilities independently, including on-ground contingency planning and last-minute problem-solving under deadline pressure.
- Demonstrated time management and leadership by keeping all departments aligned and the event on schedule from planning to execution.

Committee Member & Performer | Kachi Festival & Unity Spirit Towards Spring, UUM

Oct 2023 – Dec 2025

- Managed concurrent responsibilities across event logistics, VIP hospitality, and ceremony coordination for two university-wide cultural festivals.
- Communicated and worked directly with multiple stakeholders including event leads, performers, and university staff to keep operations running smoothly.

Performer (Saxophone) | Convocation Symphonic Band, UUM

Nov 2023 – Nov 2025

- Committed to regular rehearsal schedules while balancing academic workload, demonstrating discipline and time management.

LANGUAGES

Mandarin | Bahasa Malaysia | English

REFERENCE

Dr. Ch'ng Chee Keong

College of Arts and Sciences, School of Quantitative Sciences, Universiti Utara Malaysia

+6049286390 | chee@uum.edu.my